## Detector context

Integral + LQE

$p^T$

**SHIFT-DETR**

Final distribution $p^o$

Native decoder

Backbone + encoder



**Input image**

### 1. Shared state field

Encoder $F_8$

1 × 1 projection

Selective spatial scan

Row + column bidirectional

$h_j = a_j\, h_{j-1} + b_j$

$a_j = \exp[-\Delta_j \exp(\theta)]$

**Readout + local residual**
**Shared field S**

Computed once per image

### 2. Candidate comparison

First-layer pre-box → candidates
Three aligned sampling rays

Feature i

Feature i + 1

Query · $p_i$, $p_{i+1}$ · gap $d_i$

**Shared pair MLP**

Signed transfer score $g_i$

### 3. Bounded transport

Current side distribution p

Adjacent transfers only

$J_i = \gamma \tanh(g_i)\, p_{donor}$

Physical-distance budget
$D = \Sigma\, |J_i|\, d_i$
$\alpha = (B/T) / \max(B/T, D)$

$p'_i = p_i + \alpha(J_{i-1} - J_i)$

$\Sigma\, p'_i = 1 \qquad p'_i \geq 0$

B = ε × uncorrected final box size · T updates share the budget and reuse visual samples